

Name: _____

Date: _____

STRAIGHT-LINE GRAPHS, (PLOTting STRAIGHT-LINE GRAPHS)

LO: I can plot straight-line graphs from their equations using a table of values.

Plotting straight lines

To **plot a straight-line graph**, create a table of **x and y values** by substituting x values into the equation. Plot the points and join with a straight line.

Shortcut: You only need 2 points to draw a line, but use 3 or more to check for errors.

Quick examples

●	$y = 2x + 1$: when $x = 0$, $y = 1$	y-intercept
●	$y = 2x + 1$: when $x = 3$, $y = 7$	substitute
●	Use $x = -2, 0, 2, 4$ for a good range	include negatives
●	$y = 2x + 1$: when $x = 2$, $y = 5$	$2(2)+1 = 5$ is correct actually - but check

Key idea

Choose x values, substitute to find y, plot the coordinate pairs, join with a ruler.

$x = 0, 1, 2, 3$

substitute into $y = mx + c$

plot points

Always use a ruler for straight lines

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – positive gradient

Plot $y = 2x + 1$ for x from -2 to 3.

$x = -2: y = -3$; $x = 0: y = 1$; $x = 2: y = 5$;
 $x = 3: y = 7$

Plot $(-2, -3)$, $(0, 1)$, $(2, 5)$, $(3, 7)$

Join with a straight line

Line through $(-2, -3)$ to $(3, 7)$

Substitute each x value into the equation.

Example 2 – negative gradient

Plot $y = -x + 4$ for x from -1 to 5.

$x = -1: y = 5$; $x = 0: y = 4$; $x = 2: y = 2$;
 $x = 5: y = -1$

Plot $(-1, 5)$, $(0, 4)$, $(2, 2)$, $(5, -1)$

Line goes downhill left to right

Line through $(-1, 5)$ to $(5, -1)$

Negative gradient means the line slopes downward.

Example 3 – fractional gradient

Plot $y = \frac{1}{2}x + 2$ for x from -4 to 4.

$x = -4: y = 0$; $x = 0: y = 2$; $x = 2: y = 3$;
 $x = 4: y = 4$

Plot $(-4, 0)$, $(0, 2)$, $(2, 3)$, $(4, 4)$

Gentle upward slope

Line through $(-4, 0)$ to $(4, 4)$

Use even x values to avoid fractional y values.

Example 4 – horizontal and vertical

Plot $y = 3$ and $x = -2$.

$y = 3$ is horizontal through all points
where $y = 3$

$x = -2$ is vertical through all points

where $x = -2$

They intersect at $(-2, 3)$

**Horizontal line $y = 3$ and
vertical line $x = -2$**

Horizontal lines have equation $y = c$; vertical lines $x = a$.

YOUR TURN – PRACTICE (deliberate practice)

1. Completing tables

Complete the table for each equation ($x = -2, -1, 0, 1, 2, 3$).

- a) $y = x + 3$
- b) $y = 2x - 1$
- c) $y = -x + 2$
- d) $y = 3x + 2$

2. Identifying key features

For each equation, state the y-intercept and gradient.

- a) $y = 4x + 2$
- b) $y = -x + 5$
- c) $y = \frac{1}{3}x - 1$
- d) $y = -2x$

3. Where lines cross axes

Find where each line crosses the x-axis and y-axis.

- a) $y = x - 4$
- b) $y = 2x + 6$
- c) $y = -3x + 9$
- d) $y = \frac{1}{2}x - 3$

4. Matching equations to descriptions

State the equation that matches each description.

- a) Gradient 2, crosses y-axis at 5
- b) Horizontal line through (0, -3)
- c) Passes through origin with gradient -4
- d) Gradient $\frac{1}{2}$, y-intercept 1

5. Mixed plotting problems

Answer each question.

- a) Does (3, 10) lie on $y = 3x + 1$?
- b) Find where $y = x + 2$ and $y = 3$ intersect.
- c) A line passes through (0, -2) and (4, 6). Equation?
- d) What is the equation of the x-axis?
- e) Find the midpoint of (1, 3) and (5, 7) on a line.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) You need at least 2 points to plot a straight line ☐

Correct answer: _____

Reason: _____

b) $y = 3$ is a vertical line ☐

Correct answer: _____

Reason: _____

c) A straight line can be drawn with a ruler through plotted points ☐

Correct answer: _____

Reason: _____

d) The y-intercept is where $x = 1$ ☐

Correct answer: _____

Reason: _____

e) Substituting $x = 0$ into $y = mx + c$ gives the y-intercept ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Plot $y = 2x - 1$ and $y = -x + 5$ on the same axes for x from -1 to 4. Find their point of intersection algebraically and verify it matches your graph.

Expression: _____

Simplified: _____

b) A straight line passes through (-3, -7) and (5, 9). Find its equation and plot it. At what point does it cross the line $y = x + 2$?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

